

NORTH SOUTH UNIVERSITY

Department of Electrical and Computer Engineering

EEE 141L / ETE 141L : Electrical Circuit - I Lab



Lab Report Spring 2026

Course Section:

Faculty:

Lab Instructor:

Experiment No.

Experiment Name:

Date of Performance:

		GROUP NO.			
Name	ID	Task Assigned			
		Circuit	Simulation	Data Analysis	Lab Report

Date of Submission:

For Instructor's Use Only

SCORE:	RESUBMIT	REMARKS:	SUBMISSION			
			N	L	SL	R
	UPDATED SCORE		Date:			
	OTHER	PENALTY:	Time:			

Experiment Name: Delta-Wye Conversion

Objectives: • To perform Delta-Wye Conversion

- To verify the results with measured data.
- Solve a complex circuit using Delta-Wye conversion.

Theory:

Circuit configurations are often encountered in which the resistors do not appear to be in series or parallel. Under these conditions, it may be necessary to convert the circuit from one form to another to solve for any unknown quantities if mesh or nodal analysis is not applied.

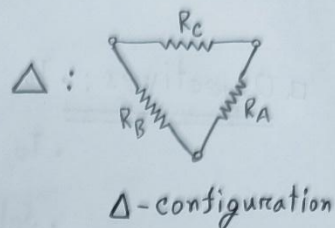
Two circuit configurations that often account for these difficulties are the Delta (Δ) - Wye (Y) configuration.

They are also referred to as the Tee (T) - Pi (π) configuration.

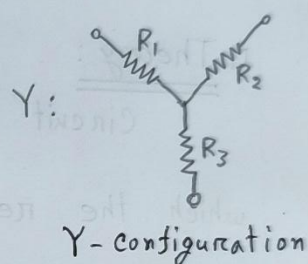
These names come from the shape of schematics.

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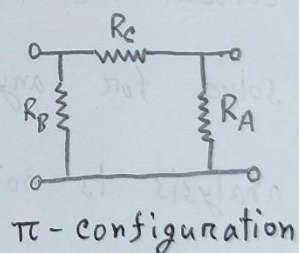
Delta Configuration: The shape of schematic is Δ or ∇



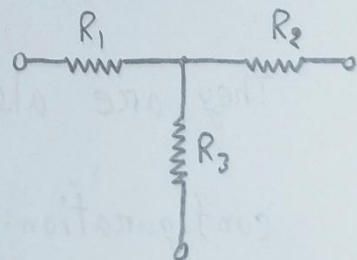
Wye Configuration: The shape of schematic is γ or λ



Pi Configuration: The shape of schematic is π .



Tee Configuration: The shape of schematic is T



T-configuration

To solve a complex circuit we may need to convert from one to another.

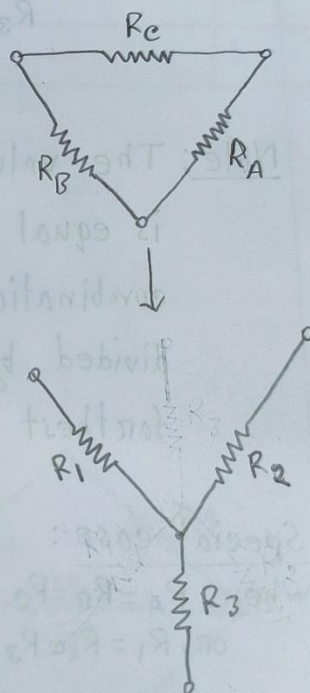
In Delta(Δ) - Wye(Y) conversion the individual resistance of each resistor or branch need to change in a way that the total equivalent resistance (R_{eq}) remains same.

Delta(Δ) to Wye(Y) conversion:

$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$



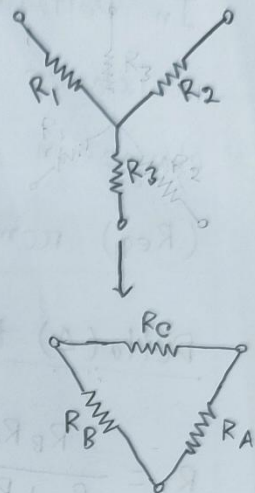
Note: Each resistor of the Wye(Y) is equal to the product of the resistors in the two closest branches of the Delta(Δ) divided by the sum of the resistors in the Delta(Δ).

Wye (Y) to Delta (Δ) conversion:

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$



Note: The value of each resistor of the Delta (Δ) is equal to the sum of the possible product combinations of the resistances of the Wye (Y) divided by the resistance of the (Wye (Y)) farthest from the resistor to be determined.

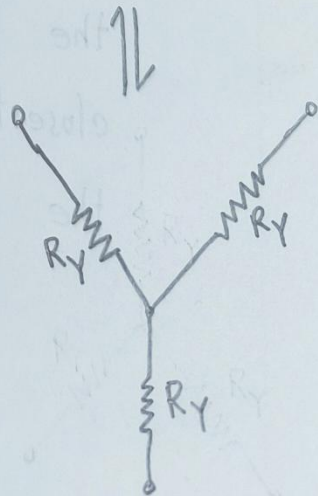
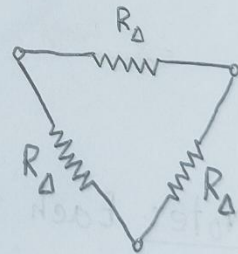
Special case:

When, $R_A = R_B = R_C$
or, $R_1 = R_2 = R_3$

$$R_Y = \frac{R_\Delta}{3}$$

and,

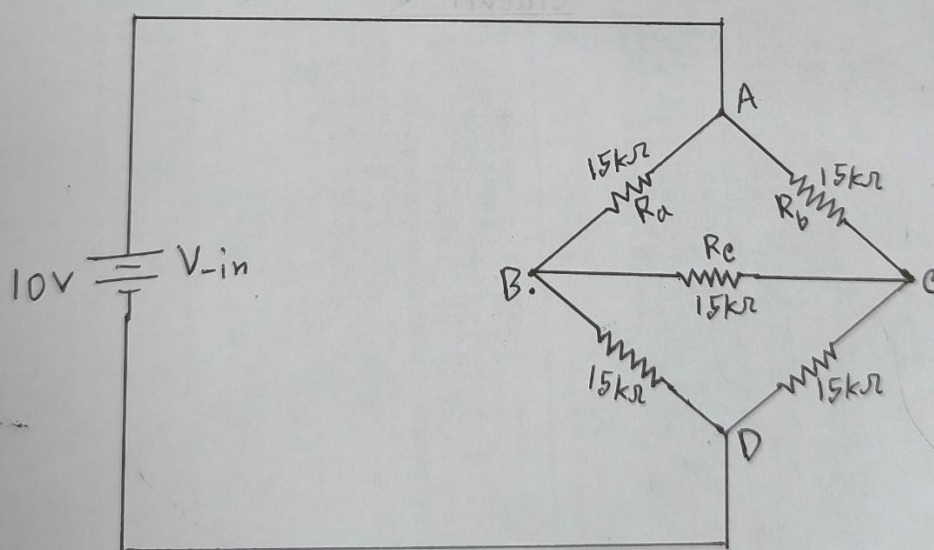
$$R_\Delta = 3 R_Y$$



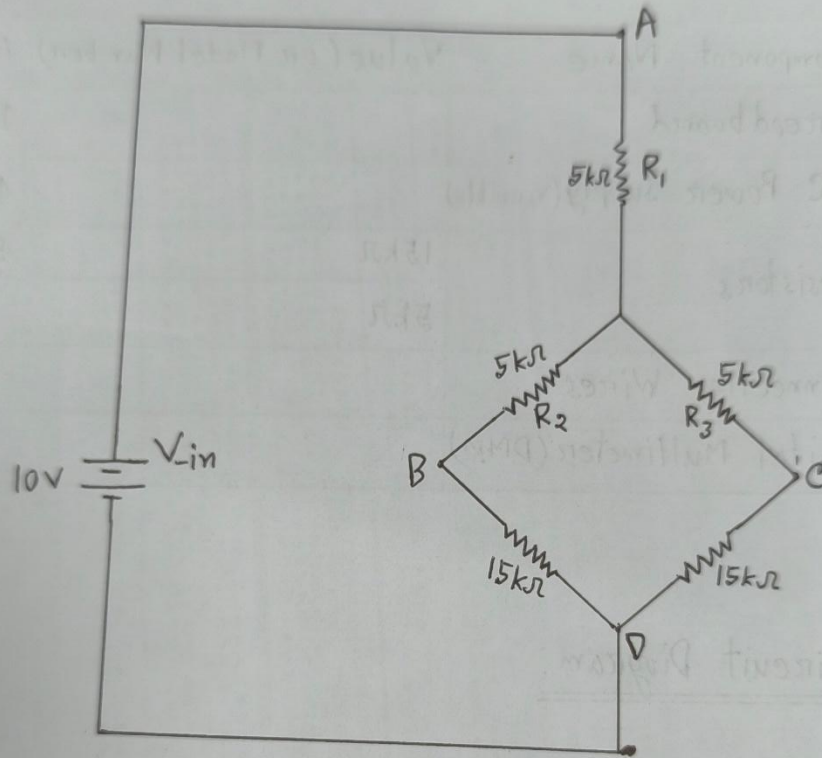
List of Equipment:

Component Name	Value (or Model Number)	Quantity
Breadboard		1
DC Power Supply (Variable)		1
Resistors	15k Ω	5
	5k Ω	3
Connecting Wires		10
Digital Multimeter (DMM)		1

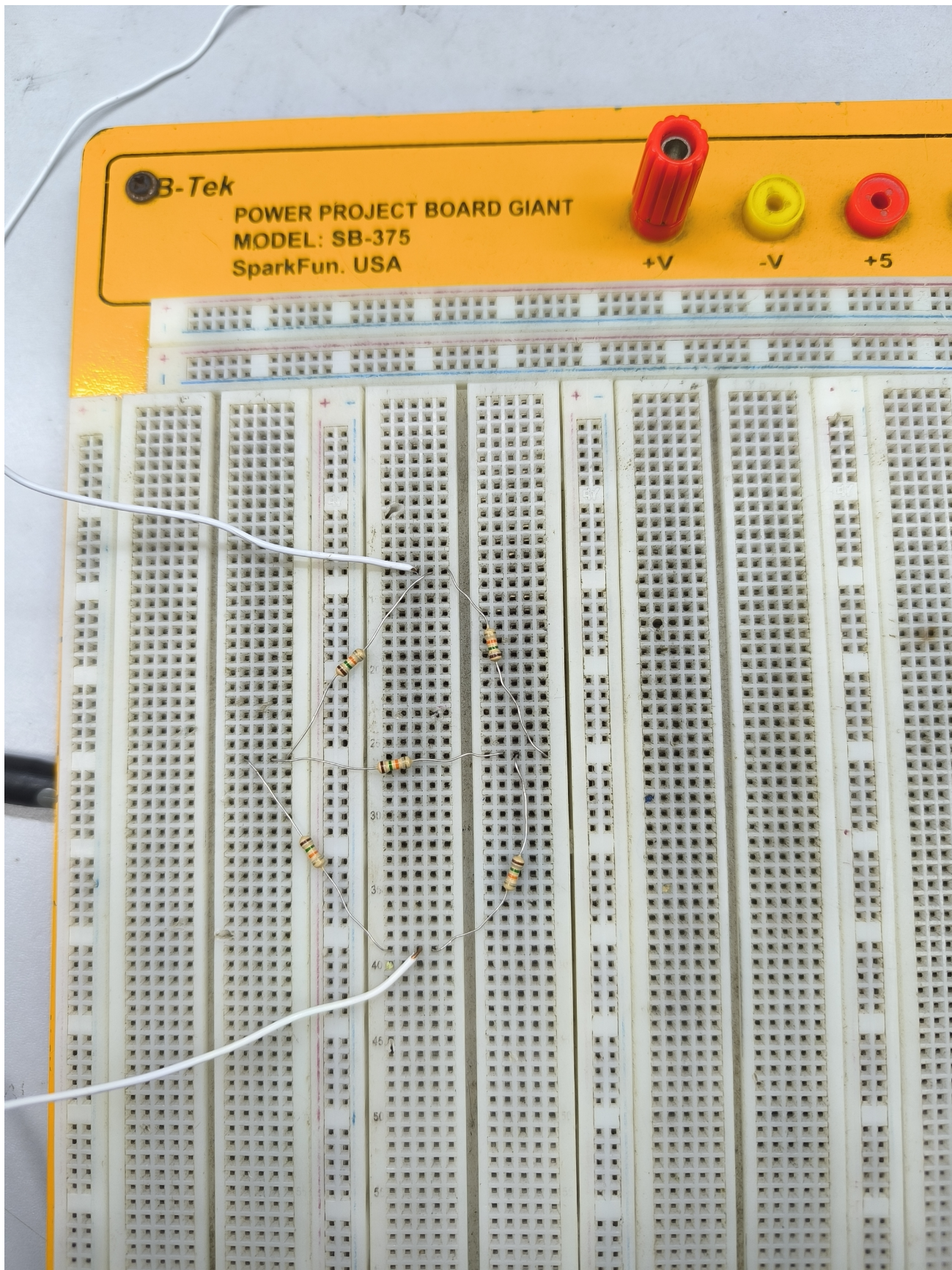
Circuit Diagram:



Circuit 1



Circuit 2



Results / Data :

Table - 1:

Resistor Label	Theoretical R	Measured R	% of Error
R _a	15 k Ω	14.46 k Ω	3.6%
R _b	15 k Ω	14.50 k Ω	3.33%
R _c	15 k Ω	14.75 k Ω	1.67%
R ₄	15 k Ω	14.55 k Ω	3%
R ₅	15 k Ω	14.40 k Ω	4.1%
R ₁	5 k Ω	5.14 k Ω	2.8%
R ₂	5 k Ω	5.10 k Ω	2%
R ₃	5 k Ω	5.14 k Ω	2.8%

$$\% \text{ of Error} = \frac{|\text{Theoretical value} - \text{Practical value}|}{\text{Theoretical value}} \times 100\%$$

$$\begin{aligned} \text{Ex. \% of error (R}_c\text{)} &= \frac{15 \text{ k}\Omega - 14.75 \text{ k}\Omega}{15 \text{ k}\Omega} \times 100\% \\ &= \frac{0.25}{15} \times 100\% \\ &= 1.67\% \end{aligned}$$

Table - 2: The total Resistance of both circuit.

Readings	Circuit 1	Circuit 2	% of Error
R _{eq}	14.73 k Ω	15.11 k Ω	2.6%

$$\begin{aligned} \% \text{ of Error} &= \frac{|\text{Circuit 1} - \text{Circuit 2}|}{\text{Circuit 1}} \times 100\% \\ &= \frac{|14.73 - 15.11|}{14.73} \times 100\% \\ &= \frac{0.38}{14.73} \times 100\% \\ &= 2.6\% \end{aligned}$$

probe in point A and negative probe in point D.

Table-3: Measure the voltages

Readings	Circuit 1	Circuit 2	% of Error
V_{AD}	10.05V	10.05V	0%
V_{BD}	5.03V	4.89V	2.8%
V_{CD}	5.02V	4.90V	2.4%
V_{AB}	5.01V	5.12V	2.19%
V_{BC}	7.7mV	10.2mV	32.5%
V_{AC}	5.02V	5.13V	2.19%

$$\% \text{ of error} = \frac{|\text{Circuit 1} - \text{Circuit 2}|}{\text{Circuit 1}} \times 100\%$$

Ex.

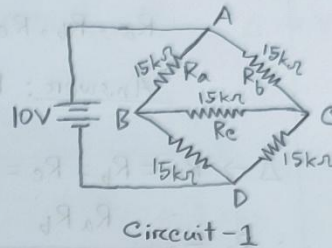
$$\text{error for } V_{CD} = \frac{(5.02 - 4.90)V}{5.02V} \times 100\% = \frac{0.12}{5.02} \times 100\% = 2.4\%$$

$$\begin{aligned} \text{error for } V_{AC} &= \frac{|5.02 - 5.13|}{5.02} \times 100\% \\ &= \frac{0.11}{5.02} \times 100\% \\ &= 2.19\% \end{aligned}$$

□ Question and Answer:

Q1. The resistors in circuit-1 are in series or in parallel combination?

Answer:



The resistors in circuit-1 are neither in simple series nor in parallel. They are connected in Delta (Δ) Configuration.

Q2. What technique would you use to find the equivalent resistance?

Answer: To find the equivalent resistance,

I would use a Delta to Wye ($\Delta \rightarrow Y$) transformation to convert the delta network into an equivalent Wye (Y) network. After the conversion, the resistors can be simplified using series and parallel combination.

To measure with DMM I would connect the positive and probe in point A and negative probe in point D.

Q3. Perform Delta-Wye conversion for ΔABC (upper portion) of circuit-1. Show all your steps to find the equivalent resistance R_1, R_2, R_3 from R_a, R_b, R_c .

Answer: Delta to Wye (Upper portion)

$$\Delta \rightarrow R_a = R_b = R_c = 15 \text{ k}\Omega$$

$$\therefore R_1 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$= \frac{15 \times 15}{15 + 15 + 15} \text{ k}\Omega$$

$$= \frac{15 \times 15}{3 \times 15} \text{ k}\Omega$$

$$= 5 \text{ k}\Omega$$

$$R_2 = \frac{R_a}{3} \quad [\because R_a = R_b = R_c]$$

$$= \frac{15 \text{ k}\Omega}{3}$$

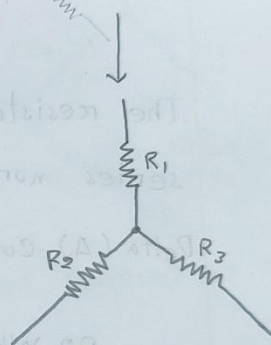
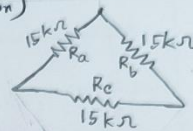
$$= 5 \text{ k}\Omega$$

$$R_3 = \frac{R_a}{3} \quad [\because R_a = R_b = R_c]$$

$$= \frac{15 \text{ k}\Omega}{3}$$

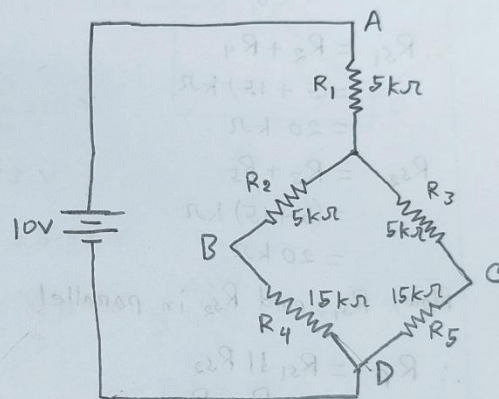
$$= 5 \text{ k}\Omega$$

$$\therefore R_1 = R_2 = R_3 = 5 \text{ k}\Omega$$



Q4. Redraw the equivalent circuit after applying the Delta-Wye conversion for ΔABC . Is it same as circuit 1?

Answer: Redrawing the circuit after $\Delta \rightarrow Y$ we get $R_1 \neq R_2 = R_3 = 5k\Omega$



Equivalent circuit of circuit-1

Yes, the equivalent circuit after Delta-Wye conversion is same as circuit-2.

Q5. Calculate R_{eq}

Answer:

$$R_1 = R_2 = R_3 = 5\text{ k}\Omega, R_4 = R_5 = 15\text{ k}\Omega$$

R_2, R_4 and R_3, R_5 are in series config,

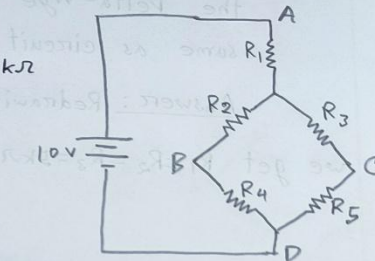
$$\begin{aligned} \therefore R_{s1} &= R_2 + R_4 \\ &= (5 + 15)\text{ k}\Omega \\ &= 20\text{ k}\Omega \end{aligned}$$

$$\begin{aligned} R_{s2} &= R_3 + R_5 \\ &= (5 + 15)\text{ k}\Omega \\ &= 20\text{ k}\Omega \end{aligned}$$

Now, R_{s1} and R_{s2} in parallel

$$\begin{aligned} \therefore R_p &= R_{s1} \parallel R_{s2} \\ &= \frac{R_{s1} \times R_{s2}}{R_{s1} + R_{s2}} \\ &= \frac{20 \times 20}{20 + 20}\text{ k}\Omega \\ &= 10\text{ k}\Omega \end{aligned}$$

$$\begin{aligned} \therefore R_{eq} &= R_1 + R_p \\ &= 5\text{ k}\Omega + 10\text{ k}\Omega \\ &= 15\text{ k}\Omega \end{aligned}$$



Q6. Calculate the voltage of R_1, R_2, R_3 .

Answer:

From Q5 we get,

$$R_2 + R_4 = R_{S1} = 20 \text{ k}\Omega$$

$$R_3 + R_5 = R_{S2} = 20 \text{ k}\Omega$$

$$R_{S1} \parallel R_{S2} = R_p = 10 \text{ k}\Omega$$

$$R_{eq} = R_1 + R_p = 15 \text{ k}\Omega$$

$$E = 10 \text{ V}$$

$$\therefore V_{R_1} = \frac{R_1}{R_{eq}} \times E = \frac{5 \text{ k}\Omega}{15 \text{ k}\Omega} \times 10 \text{ V}$$

$$= 3.33 \text{ V}$$

$$V_{ED} = E - V_{R_1} = 10 - 3.33 = 6.67 \text{ V}$$

$$\therefore V_{R_2} = V_{EB} = \frac{R_2}{R_2 + R_4} \times V_{ED} = \frac{5 \text{ k}\Omega}{(5 + 15) \text{ k}\Omega} \times 6.67 \text{ V}$$

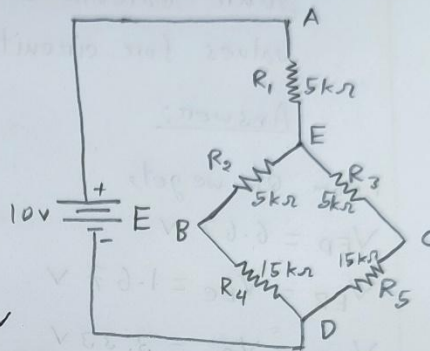
$$= 1.67 \text{ V}$$

$$\therefore V_{R_3} = V_{EC} = \frac{R_3}{R_3 + R_5} \times V_{ED} = \frac{5 \text{ k}\Omega}{(5 + 15) \text{ k}\Omega} \times 6.67 \text{ V}$$

$$= 1.67 \text{ V}$$

$$\therefore V_{R_1} = 3.33 \text{ V}$$

$$V_{R_2} = V_{R_3} = 1.67 \text{ V}$$



Q7. Calculate V_{AB} , V_{BC} , V_{AC} and V_{AD} , V_{BD} , V_{ED} . Do your calculated values match the measured values for circuit 2? Find the % error.

Answer:

From Q6 we get,

$$V_{ED} = 6.67 \text{ V}$$

$$V_{EB} = V_{EC} = 1.67 \text{ V}$$

$$V_{AE} = V_{R_1} = 3.33 \text{ V}$$

$$\therefore V_{AB} = V_{AE} + V_{EB} = (3.33 + 1.67) \text{ V} = 5 \text{ V}$$

$$\begin{aligned} \therefore V_{BC} &= V_{BE} + V_{EC} = -V_{EB} + V_{EC} \\ &= (-1.67 + 1.67) \text{ V} \\ &= 0 \text{ V} \end{aligned}$$

$$\begin{aligned} \therefore V_{AC} &= V_{AE} + V_{EC} = (3.33 + 1.67) \text{ V} \\ &= 5 \text{ V} \end{aligned}$$

$$\therefore V_{AD} = E = 10 \text{ V}$$

$$\therefore V_{BD} = E - V_{AD} = (10 - 5) \text{ V} = 5 \text{ V}$$

$$\therefore V_{ED} = E - V_{AE} = (10 - 5) \text{ V} = 5 \text{ V}$$

$$\therefore V_{AB} = V_{AC} = V_{BD} = V_{ED} = 5 \text{ V}$$

$$V_{BC} = 0 \text{ V}$$

$$V_{AD} = 10 \text{ V}$$

From table-2:

We get the measured values for circuit-2:

$$V_{AD} = 10.05 \text{ V}, V_{BD} = 4.89 \text{ V}, V_{CD} = 4.90 \text{ V}$$

$$V_{AB} = 5.12 \text{ V}, V_{BC} = 10.2 \text{ mV}, V_{AC} = 5.13 \text{ V}$$

% of Errors,

$$\text{for } V_{AD} = \frac{|T.V - P.V|}{T.V} \times 100\%$$

$$= \frac{|10 - 10.05|}{10} \times 100\%$$

$$= 0.5\%$$

$$\text{for } V_{BD} = \frac{5 - 4.89}{5} \times 100\% = 2.2\%$$

$$\text{for } V_{CD} = \frac{5 - 4.90}{5} \times 100\% = 2\%$$

$$\text{for } V_{AB} = \frac{|5 - 5.12|}{5} \times 100\% = 2.4\%$$

$$\text{for } V_{BC} = \frac{0 - 0.0102}{0} \times 100\% = \text{undefined}$$

$\therefore$ Absolute error for V_{BC} is 10.2 mV

$$\text{for } V_{AC} = \frac{|5 - 5.13|}{5} \times 100\%$$

$$= 2.6\%$$

Q8. Using Table-2, analyze wheather Circuit-2 is equivalent to Circuit 1 ? Was Delta-Wye conversion successful?

Answer:

Table-2:

Readings	Circuit 1	Circuit 2	% of Error
Req	14.73 k Ω	15.11 k Ω	2.6%

From table-2 we see:

Req of Circuit 1 is 14.73 k Ω

and Req of Circuit 2 is 15.11 k Ω

So, Req in both circuit is almost simillar and the

% of error is only 2.6%. And the error is mainly the error of measurement.

It proves that the Circuit-2 is equivalent to Circuit 1.

$\therefore$ The Delta(Δ) - Wye(Y) conversion was successful.

Result Analysis and Discussion:

In this experiment, we focused on Delta-Wye transformation the fundamental technique used in analyzing complex electrical circuit. In this lab session we have explored the concept of Delta(Δ) - Wye(Y) conversion, the conversion formula's and its practical application.

We transformed a delta(Δ) network to wye(Y) network in a complex circuit. And from the both circuit we found the ^{almost} same equivalent resistance after measurement $R_{eq}(C1) = 14.73 k\Omega$ and $R_{eq}(C2) = 15.11 k\Omega$. This measured value is also similar to the theoretical value of $R_{eq}(\text{Theoretical}) = 15 k\Omega$ with some error.

Comparing the measured values in table 1, table 2 and table 3 with theoretical values, We got right data and the small error is ignorable.

The voltage readings of V_{AD} , V_{BD} , V_{CD} , V_{AB} , V_{BC} , V_{AC} from both the circuit 1 and 2 also proves that the Delta(Δ) - Wye(Y) conversion

was successful, as we got similar values in Circuit-2 compared to Circuit-1.

Circuit 1, $V_{AD} = 10.05 \text{ V}$

Circuit 2, $V_{AD} = 10.05 \text{ V}$

$V_{BD} = 5.03 \text{ V}$

$V_{BD} = 4.89 \text{ V}$

$V_{CD} = 5.02 \text{ V}$

$V_{CD} = 4.90 \text{ V}$

$V_{AB} = 5.01 \text{ V}$

$V_{AB} = 5.12 \text{ V}$

$V_{BC} = 7.7 \text{ mV}$

$V_{BC} = 10.2 \text{ mV}$

$V_{AC} = 5.02 \text{ V}$

$V_{AC} = 5.13 \text{ V}$

As per the concept of bridge network the voltage ~~netw~~ reading of V_{BC} should have been 0. ($V_{BC} = 0 \text{ V}$)

But we got a small voltage 7.7 mV (Circuit 1) and 10.2 mV (Circuit-2).

The bridge is balanced. But the reason of getting small voltage reading in DMM, is the internal resistance of the DMM itself.

Some percentage error for experimental readings can be because of loose connection of wire, resistor tolerance, and the accuracy limitation of the equipment throughout the experiment.

So, the results satisfies our theoretical concept of Delta(A) - Wye(Y) Conversion.

□ Conclusion: The experiment was successful. Analyzing the result of the experiment, which satisfies the concept of Delta(Δ)-Wye(Y) conversion; we can conclude that, Delta(Δ)-Wye(Y) transformation is really helpful and proven method to solve a complex circuit where simple series-parallel network is hard or impossible to determine.

□ Reference:

1. Lab Manual
2. Introductory Circuit Analysis, Robert L. Boylestad
3. Fundamentals of Electrical Circuits, Charles K Alexander
4. Datasheet
5. Slides from lab class

□ Attachment:

1. Printed image of practical breadboard circuit setup, Circuit-1 and Circuit-2
2. Printed copies of Multisim Simulations.
3. Signed Datasheet.

Data Collection:

Lab 5:

Group No. 3

Instructor's Signature

Audipto 2/4/26

Table 1:

Resistor Label	Theoretical R	Measured R	% of Error
R_a	15k Ω	14.46 k Ω	3.6%
R_b	15k Ω	14.50 k Ω	3.33%
R_c	15k Ω	14.75 k Ω	1.67%
R_d	15k Ω	14.55 k Ω	3%
R_s	15k Ω	14.40 k Ω	4%
R_x	5k Ω	5.14 k Ω	2.8%
R_y	5k Ω	5.10 k Ω	2%
R_z	5k Ω	5.14 k Ω	2.8%

Table 2: Measure the total resistance of both circuits

Readings	Circuit 1	Circuit 2	% of Error
R_{total}	14.73 k Ω	15.11 k Ω	2.6%

Table 3: Measure the voltages

Readings	Circuit 1	Circuit 2	% of Error
V_{AD}	10.05 V	10.05 V	0%
V_{BD}	5.03 V	4.89 V	2.8%
V_{CD}	5.02 V	4.90 V	2.58%
V_{AB}	5.01 V	5.12 V	2.19%
V_{BC}	17.7 mV	13.2 mV	25%
V_{AC}	5.02 V	5.13 V	2.19%

Questions:

1. Why do we need to do Delta – Wye conversion?
2. What are balanced and unbalanced load in electrical circuits?
3. Are the resistors in Circuit 1 in series or in parallel combination?
4. Perform Delta-Wye conversion for ΔABC (upper portion) of circuit 1. Show all your steps to find the equivalent resistance R_x , R_y , R_z from R_a , R_b , R_c .
5. Redraw the equivalent the circuit after applying the Delta-Wye conversion for ΔABC . Is it same as circuit 2?
6. Calculate R_{total} .
7. Calculate V_{AB} , V_{BC} , V_{AC} and V_{AD} , V_{BD} , V_{CD} in circuit 2. Do your calculated values match the measured values for circuit 2? Find the % of Error.
8. Using Table 3, analyze whether Circuit 2 is equivalent to Circuit 1? Was Delta-Wye conversion successful?